

EXAMEN FINAL

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1. Configuramos variables

```
export PROJECT_ID="imperial-time-481402-q1"
export REGION="us-central1"
export BUCKET_NAME="bi-supermarket-$(date +%s)"
export DATASET_NAME="retail_dw"
export CSV_FILE="SuperMarket Analysis.csv"
```

gcloud config set project \$PROJECT_ID

gcloud config set compute/region \$REGION

```
paololaricocruz54@cloudshell:~ (imperial-time-481402-q1) $ # --- VARIABLES ---
export PROJECT_ID="imperial-time-481402-q1"
export REGION="us-central1"
export BUCKET_NAME="bi-supermarket-$(date +%s)" # Nombre único
export DATASET_NAME="retail_dw"
export CSV_FILE="SuperMarket Analysis.csv"
```

2. Habilitamos los servicios

```
gcloud services enable \
  storage-component.googleapis.com \
  dataproc.googleapis.com \
  bigquery.googleapis.com \
  artifactregistry.googleapis.com
```

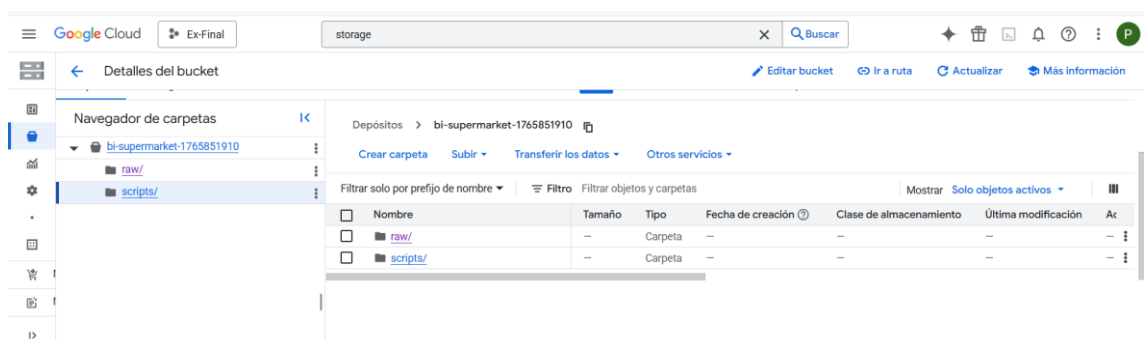
```
paololaricocruz54@cloudshell:~ (imperial-time-481402-q1) $ gcloud services enable \
  storage-component.googleapis.com \
  dataproc.googleapis.com \
  bigquery.googleapis.com \
  artifactregistry.googleapis.com

Operation "operations/acf.p2-760509638482-481d1555-f873-4f78-ad8e-78a290b793de" finished successfully.
paololaricocruz54@cloudshell:~ (imperial-time-481402-q1) $
```

3. Creamos el bucket

```
gsutil mb -l $REGION gs://$BUCKET_NAME
```

```
paololaricocruz54@cloudshell:~ (imperial-time-481402-q1) $ gsutil mb -l $REGION gs://$BUCKET_NAME
Creating gs://bi-supermarket-1765851910/...
paololaricocruz54@cloudshell:~ (imperial-time-481402-q1) $ touch .keep
```



4. Creamos la estructura de carpetas

```
touch .keep
```

```
gsutil cp .keep gs://$BUCKET_NAME/raw/.keep
```

```
gsutil cp .keep gs://$BUCKET_NAME/scripts/.keep
rm .keep
```

```
paololaricocruz54@cloudshell:~ (imperial-time-481402-q1)$ touch .keep
gsutil cp .keep gs://$BUCKET_NAME/raw/.keep
gsutil cp .keep gs://$BUCKET_NAME/scripts/.keep
rm .keep
```

5. Creamos el dataset en bigquery

```
bq mk --location=$REGION --dataset $PROJECT_ID:$DATASET_NAME
```

```
paololaricocruz54@cloudshell:~/proyecto_supermarket (singular-cache-481401-j2)$ bq mk --location=$REGION --force bi_supermarket_db || true
Dataset 'singular-cache-481401-j2:bi_supermarket_db' successfully created.
```

6. Subimos el archivo a la carpeta raw

7. gsutil cp "\$CSV_FILE" "gs://\$BUCKET_NAME/raw/\$CSV_FILE"

```
paololaricocruz54@cloudshell:~/proyecto_supermarket (singular-cache-481401-j2)$ gsutil cp ~/supermarket_analysis.csv gs://$BUCKET_NAME/bronze/raw/
Copying file:///home/paololaricocruz54/supermarket_analysis.csv [Content-Type=text/csv]...
/ [1 files][137.0 KiB/137.0 KiB]
Operation completed over 1 objects/137.0 KiB.
```

8. Ahora creamos el scripts para la limpieza simple

```
cat <<EOF > etl_supermarket_job.py
```

```
import sys
```

```
from pyspark.sql import SparkSession
```

```
from pyspark.sql.functions import col, upper, trim, current_timestamp
```

```
from pyspark.sql.types import DoubleType, IntegerType
```

```
project_id = "$PROJECT_ID"
```

```
bucket_name = "$BUCKET_NAME"
```

```
dataset_name = "$DATASET_NAME"
```

```
spark = SparkSession.builder.appName("ETL-Supermarket").getOrCreate()
```

```
# A. LEER
```

```
df_raw = spark.read.option("header", True).option("inferSchema",
True).csv(f"gs://{bucket_name}/raw/*.csv")
```

```
# B. TRANSFORMAR
```

```
df_curated = df_raw \
```

```
    .withColumnRenamed("Invoice ID", "invoice_id") \
```

```
    .withColumnRenamed("Customer type", "customer_type") \
```

```
    .withColumnRenamed("Product line", "product_line") \
```

```
    .withColumnRenamed("Unit price", "unit_price") \
```

```
    .withColumnRenamed("Tax 5%", "tax_5_percent") \
```

```
    .withColumnRenamed("Gross income", "gross_income") \
```

```
    .withColumn("unit_price", col("unit_price").cast(DoubleType())) \
```

```
    .withColumn("Quantity", col("Quantity").cast(IntegerType())) \
```

```
    .withColumn("Sales", col("Sales").cast(DoubleType())) \
```

```
    .withColumn("gross_income", col("gross_income").cast(DoubleType())) \
```

```
    .withColumn("Rating", col("Rating").cast(DoubleType())) \
```

```
    .withColumn("Branch", upper(trim(col("Branch")))) \
```

```
    .withColumn("City", upper(trim(col("City")))) \
```

```
.withColumn("customer_type", upper(trim(col("customer_type")))) \
.withColumn("product_line", upper(trim(col("product_line")))) \
.withColumn("Payment", upper(trim(col("Payment")))) \
.na.drop(subset=["invoice_id"])
```

C. ESCRIBIR

```
df_curated.write \
    .format("bigquery") \
    .option("table", f"{project_id}.{dataset_name}.tbl_supermarket_curated") \
    .option("temporaryGcsBucket", bucket_name) \
    .mode("overwrite") \
    .save()
```

EOF

```
paololaricocruz54@cloudshell:~/proyecto_supermarket (singular-cache-481401-j2)$
cat <EOF> etl_supermarket.py
import sys
from pyspark.sql import SparkSession
from pyspark.sql.functions import col, upper, trim, current_timestamp
from pyspark.sql.types import DoubleType, IntegerType

project_id = "$PROJECT_ID"
bucket_name = "$BUCKET_NAME"
dataset_name = "$DATASET_NAME"

spark = SparkSession.builder.appName("ETL-Supermarket").getOrCreate()

# A. LEER
df_raw = spark.read.option("header", True).option("inferSchema", True).csv(f"gs://{bucket_name}/raw/*.csv")

# B. TRANSFORMAR
df_curated = df_raw \
    .withColumnRenamed("Invoice ID", "invoice_id") \
    .withColumnRenamed("Customer type", "customer_type") \
    .withColumnRenamed("Product line", "product_line") \
    .withColumnRenamed("Unit price", "unit_price") \
    .withColumnRenamed("Tax 5%", "tax_5_percent") \
    .withColumnRenamed("Gross income", "gross_income") \
    .save(f"gs://{bucket_name}/tbl_supermarket_curated") \
EOF
paololaricocruz54@cloudshell:~/proyecto_supermarket (singular-cache-481401-j2)$ nano etl_supermarket.py
```

9. Guardamos el script en la carpeta correspondiente

```
gsutil cp etl_supermarket_job.py
```

```
"gs://$BUCKET_NAME/scripts/etl_supermarket_job.py"
```

```
paololaricocruz54@cloudshell:~/proyecto_supermarket (singular-cache-481401-j2)$ gsutil cp etl_supermarket.py "gs://$BUCKET_NAME/scripts/etl_supermarket.py"
Copying file://etl_supermarket.py [Content-Type=text/x-python]...
/ [1 files] [ 1.7 KiB/ 1.7 KiB]
Operation completed over 1 objects/1.7 KiB.
```

10. Ejecutamos el job mediante comando CLI

```
gcloud dataproc batches submit pyspark \
```

```
gs://$BUCKET_NAME/scripts/etl_supermarket_job.py \
```

```
--region=$REGION \
```

```
--deps-bucket=gs://$BUCKET_NAME \
```

```
--jars=gs://spark-lib/bigquery/spark-bigquery-latest_2.12.jar \
```

```
--version=2.1 \
```

```
--subnet=default
```

```
{
  "formatOptions": {
    "format": "PARQUET",
    "nullMarker": null,
    "maxBadRecords": null,
    "schema": {
      "fields": [
        {
          "name": "invoice_id",
          "type": "STRING",
          "mode": "NULLABLE",
          "description": null,
          "policyTags": null,
          "maxLength": null,
          "scale": null,
          "precision": null,
          "defaultValueExpression": null
        },
        {
          "name": "Branch",
          "type": "STRING",
          "mode": "NULLABLE",
          "description": null,
          "policyTags": null,
          "maxLength": null,
          "scale": null,
          "precision": null,
          "defaultValueExpression": null
        },
        {
          "name": "City",
          "type": "STRING",
          "mode": "NULLABLE",
          "description": null,
          "policyTags": null,
          "maxLength": null,
          "scale": null,
          "precision": null,
          "defaultValueExpression": null
        },
        {
          "name": "Customer type",
          "type": "STRING",
          "mode": "NULLABLE",
          "description": null,
          "policyTags": null,
          "maxLength": null,
          "scale": null,
          "precision": null,
          "defaultValueExpression": null
        },
        {
          "name": "product_line",
          "type": "STRING",
          "mode": "NULLABLE",
          "description": null,
          "policyTags": null,
          "maxLength": null,
          "scale": null,
          "precision": null,
          "defaultValueExpression": null
        },
        {
          "name": "unit_price",
          "type": "FLOAT",
          "mode": "NULLABLE",
          "description": null,
          "policyTags": null,
          "maxLength": null,
          "scale": null,
          "precision": null,
          "defaultValueExpression": null
        },
        {
          "name": "tax_5_percent",
          "type": "FLOAT",
          "mode": "NULLABLE",
          "description": null,
          "policyTags": null,
          "maxLength": null,
          "scale": null,
          "precision": null,
          "defaultValueExpression": null
        },
        {
          "name": "gross_income",
          "type": "FLOAT",
          "mode": "NULLABLE",
          "description": null,
          "policyTags": null,
          "maxLength": null,
          "scale": null,
          "precision": null,
          "defaultValueExpression": null
        },
        {
          "name": "Rating",
          "type": "FLOAT",
          "mode": "NULLABLE",
          "description": null,
          "policyTags": null,
          "maxLength": null,
          "scale": null,
          "precision": null,
          "defaultValueExpression": null
        },
        {
          "name": "Payment",
          "type": "STRING",
          "mode": "NULLABLE",
          "description": null,
          "policyTags": null,
          "maxLength": null,
          "scale": null,
          "precision": null,
          "defaultValueExpression": null
        },
        {
          "name": "margin_percentage",
          "type": "FLOAT",
          "mode": "NULLABLE",
          "description": null,
          "policyTags": null,
          "maxLength": null,
          "scale": null,
          "precision": null,
          "defaultValueExpression": null
        },
        {
          "name": "gross_margin_percentage",
          "type": "FLOAT",
          "mode": "NULLABLE",
          "description": null,
          "policyTags": null,
          "maxLength": null,
          "scale": null,
          "precision": null,
          "defaultValueExpression": null
        },
        {
          "name": "cogs",
          "type": "FLOAT",
          "mode": "NULLABLE",
          "description": null,
          "policyTags": null,
          "maxLength": null,
          "scale": null,
          "precision": null,
          "defaultValueExpression": null
        }
      ]
    }
  },
  "ignoreUnknownValue": null,
  "sourceUri": "gs://bi-supermarket-1765851910/spark-bigquery-app-20251216023355-0000-d4b60181-4d00-43a0-5d32-7d3bd429f6b1/part-00000-91b36b40-592a-4860-aa4e-925ad562f00-c000.snappy.parquet",
  "schemaUpdateOptions": null,
  "autodetect": null,
  "timePartitioning": null,
  "clustering": null,
  "userDefinedLogicalTypes": null,
  "labels": null,
  "jobTimeout": null,
  "rangePartitioning": null,
  "hivePartitioningOptions": null,
  "referenceFileSchemaUri": null,
  "jobId": "j_0b1d7project=imperial-time-481402-gl_1db6d053d33-927e-4d4b-bab8-1c6dd1fd0b89_location=us-central1",
  "25/12/16 02:34:11 INFO BigQueryClient: Done loading to Imperial-time-481402-gl.retail_dw.tbl_supermarket_curated. jobId: JobId{project=imperial-time-481402-gl_job=6d053d33-927e-4d4b-bab8-1c6dd1fd0b89_location=us-central1}
Batch [afe06fd3ca9444a9886ae5bed9998ad] Finished.
metadata:
  "type": "type.googleapis.com/google.cloud.dataproc.v1.BatchOperationMetadata",
  "batch": {
    "project": "imperial-time-481402-gl/locations/us-central1/batches/afe06fd3ca9444a9886ae5bed9998ad",
    "batchUuid": "43824470-3d6c-445f-92ac-6251105133f6",
    "createTime": "2025-12-16T02:31:56.677704Z",
    "description": "Batch",
    "labels": {
      "goog-dataproc-batch-id": "afe06fd3ca9444a9886ae5bed9998ad",
      "goog-dataproc-batch-uuid": "43824470-3d6c-445f-92ac-6251105133f6",
      "goog-dataproc-driver-source-uuid": "batch-43824470-3d6c-445f-92ac-6251105133f6",
      "goog-dataproc-location": "us-central1",
      "operationType": "BATCH"
    },
    "name": "projects/imperial-time-481402-gl/regions/us-central1/operations/f8e0a6ae-7d61-3063-a515-495127ec7bee"
  }
}
```

```

11. Construimos el modelo estrella, cargamos las dimensiones y la tabla de hecho
bq query --use_legacy_sql=false \
"CREATE OR REPLACE TABLE \ ` $PROJECT_ID.$DATASET_NAME.dim_sucursal\`
AS
SELECT DISTINCT ABS(FARM_FINGERPRINT(Branch)) as id_sucursal, Branch, City
FROM \ ` $PROJECT_ID.$DATASET_NAME.tbl_supermarket_curated\`;
CREATE OR REPLACE TABLE \ ` $PROJECT_ID.$DATASET_NAME.dim_producto\` AS
SELECT DISTINCT ABS(FARM_FINGERPRINT(product_line)) as id_producto,
product_line FROM \ ` $PROJECT_ID.$DATASET_NAME.tbl_supermarket_curated\`;
CREATE OR REPLACE TABLE \ ` $PROJECT_ID.$DATASET_NAME.dim_cliente\` AS
SELECT DISTINCT ABS(FARM_FINGERPRINT(CONCAT(customer_type, Gender)))
as id_cliente, customer_type, Gender FROM
\ ` $PROJECT_ID.$DATASET_NAME.tbl_supermarket_curated\`;
CREATE OR REPLACE TABLE \ ` $PROJECT_ID.$DATASET_NAME.dim_pago\` AS
SELECT DISTINCT ABS(FARM_FINGERPRINT(Payment)) as id_pago, Payment FROM
\ ` $PROJECT_ID.$DATASET_NAME.tbl_supermarket_curated\`;
CREATE OR REPLACE TABLE \ ` $PROJECT_ID.$DATASET_NAME.dim_tiempo\` AS
SELECT DISTINCT ABS(FARM_FINGERPRINT(CAST(Date AS STRING))) as
id_tiempo, CAST(Date AS DATE) as fecha, EXTRACT(MONTH FROM CAST(Date AS
DATE)) as mes FROM
\ ` $PROJECT_ID.$DATASET_NAME.tbl_supermarket_curated\`;"

```

```

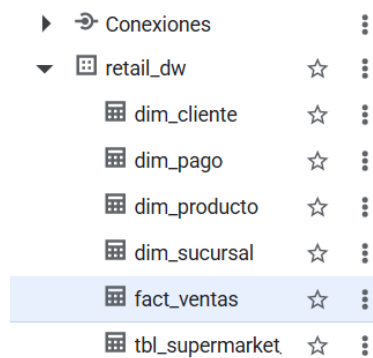
bq query --use_legacy_sql=false \
"CREATE OR REPLACE TABLE \ ` $PROJECT_ID.$DATASET_NAME.fact_ventas\` AS
SELECT
  ABS(FARM_FINGERPRINT(Branch)) as id_sucursal,
  ABS(FARM_FINGERPRINT(product_line)) as id_producto,
  ABS(FARM_FINGERPRINT(CONCAT(customer_type, Gender))) as id_cliente,
  ABS(FARM_FINGERPRINT(Payment)) as id_pago,
  ABS(FARM_FINGERPRINT(CAST(Date AS STRING))) as id_tiempo,
  invoice_id, Sales as total, tax_5_percent as tax, gross_income, Quantity as
quantity
FROM \ ` $PROJECT_ID.$DATASET_NAME.tbl_supermarket_curated\`;"

```

```

paololaricocruz54@cloudshell:~ (imperial-time-481402-ql) $ bq query --use_legacy_sql=false \
"CREATE OR REPLACE TABLE \ ` $PROJECT_ID.$DATASET_NAME.dim_sucursal\` AS
SELECT DISTINCT ABS(FARM_FINGERPRINT(Branch)) as id_sucursal, Branch, City FROM \ ` $PROJECT_ID.$DATASET_NAME.tbl_supermarket_curated\`;
CREATE OR REPLACE TABLE \ ` $PROJECT_ID.$DATASET_NAME.dim_producto\` AS
SELECT DISTINCT ABS(FARM_FINGERPRINT(product_line)) as id_producto, product_line FROM \ ` $PROJECT_ID.$DATASET_NAME.tbl_supermarket_curated\`;
CREATE OR REPLACE TABLE \ ` $PROJECT_ID.$DATASET_NAME.dim_cliente\` AS
SELECT DISTINCT ABS(FARM_FINGERPRINT(CONCAT(customer_type, Gender))) as id_cliente, customer_type, Gender FROM \ ` $PROJECT_ID.$DATASET_NAME.tbl_supermarket_curated\`;
CREATE OR REPLACE TABLE \ ` $PROJECT_ID.$DATASET_NAME.dim_pago\` AS
SELECT DISTINCT ABS(FARM_FINGERPRINT(Payment)) as id_pago, Payment FROM \ ` $PROJECT_ID.$DATASET_NAME.tbl_supermarket_curated\`;
CREATE OR REPLACE TABLE \ ` $PROJECT_ID.$DATASET_NAME.dim_tiempo\` AS
SELECT DISTINCT ABS(FARM_FINGERPRINT(CAST(Date AS STRING))) as id_tiempo, CAST(Date AS DATE) as fecha, EXTRACT(MONTH FROM CAST(Date AS DATE)) as mes FROM \ ` $PROJECT_ID.$DATASET_NAME.tbl_super
echo " Modelo Estrella creado."E.tbl_supermarket_curated\`; "income, Quantity as quantity
Waiting on bqjob_f7a2004977a69ab75_0000019b2504e7d9_1 ... (7s) Current status: DONE
Error in query string: Error processing job 'imperial-time-481402-ql:bqjob_f7a2004977a69ab75_0000019b2504e7d9_1': Query error: Invalid date: '2/8/2019' at [9:1]
Waiting on bqjob_f7bfc8a7ab012204_0000019b250515d1_1 ... (1s) Current status: DONE
Created imperial-time-481402-ql:retail_dw.fact_ventas
Modelo Estrella creado.
paololaricocruz54@cloudshell:~ (imperial-time-481402-ql) $

```



12. Verificamos resultados

fact_ventas

☆

Q

Consulta

🔗

Abrir en ▾

👤

Compartir ▾

📄

Copiar

🕒

Instantánea

🗑️

Borrar

📤

Exportación

🔄

Actualiz

Esquema

Detalles

Vista previa

Explorador de tablas

Vista previa

Estadísticas

Linaje

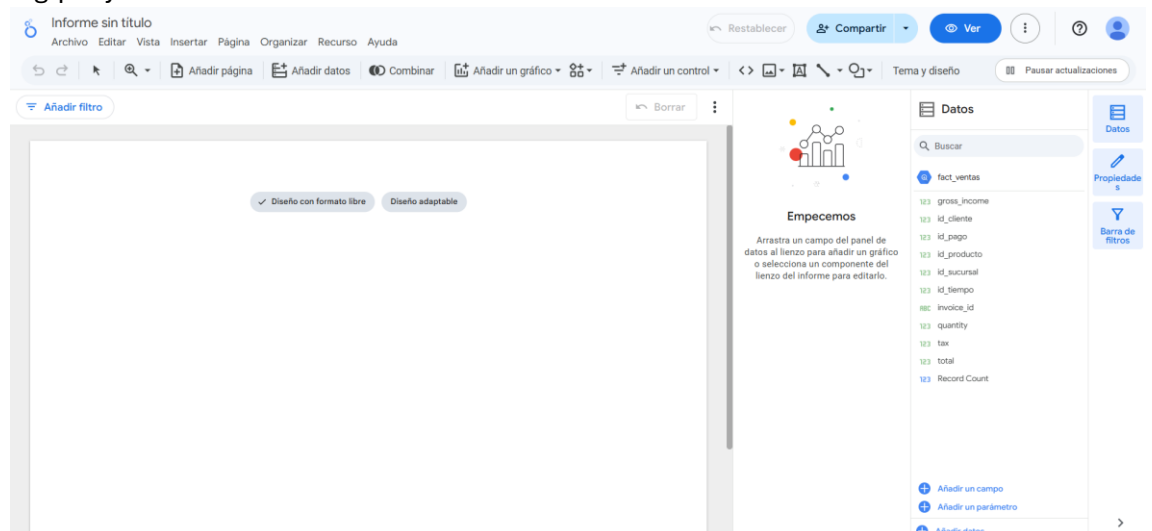
Perfil de datos

Calidad de los datos

Fila	id_sucursal	id_producto	id_cliente	id_pago	id_tiempo	invoice_id	total	tax	gross_income	quantity
1	2599702706...	1550746522...	4323410103...	7609322075...	1061032265...	347-56-2442	69.111	3.291	3.291	
2	2599702706...	1550746522...	5908857420...	8066885345...	7948075386...	856-22-8149	26.5545	1.2645	1.2645	
3	2599702706...	1550746522...	4323410103...	8709920426...	8709594100...	802-43-8934	19.194	0.914	0.914	
4	6903822250...	1550746522...	4323410103...	7609322075...	1448994460...	556-72-8512	24.108	1.148	1.148	
5	2599702706...	1550746522...	4323410103...	8066885345...	2077222351...	400-60-7251	77.7735	3.7035	3.7035	
6	6903822250...	1550746522...	8938443163...	7609322075...	4847218641...	593-95-4461	78.603	3.743	3.743	
7	6903822250...	1550746522...	2522685395...	7609322075...	9087982230...	779-06-0012	93.0405	4.4305	4.4305	
8	2599702706...	1550746522...	4323410103...	8066885345...	4683090995...	643-38-7867	102.837	4.897	4.897	
9	6903822250...	1550746522...	2522685395...	7609322075...	6636124631...	443-60-9639	63.9135	3.0435	3.0435	
10	6903822250...	1550746522...	2522685395...	8066885345...	2536340420...	848-95-6252	90.5835	4.3135	4.3135	
11	5689455709...	1550746522...	2522685395...	8066885345...	1071501163...	280-35-5823	91.056	4.336	4.336	
12	6903822250...	1550746522...	4323410103...	8709920426...	7748711136...	523-38-0215	38.85	1.85	1.85	
13	6903822250...	1550746522...	8938443163...	8066885345...	3796937735...	236-86-3015	14.679	0.699	0.699	

13. Esta ok, hacemos conexión con Looker

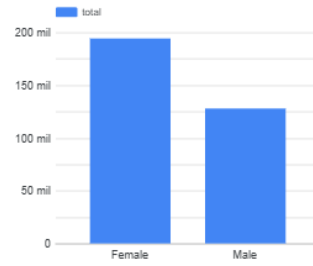
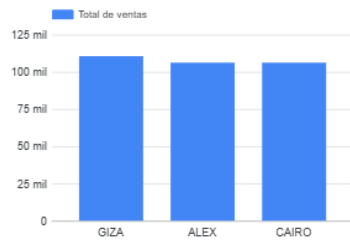
Cargamos nuestra tabla de la capa oro a looker, se puede por conexión directa con bigquery



Link del dashboard: <https://lookerstudio.google.com/reporting/31ff11f2-382b-4560-82cb-634f9ceda19b>

14. Realizamos el dashboard

DASHBOARD GENERAL



Total de ventas
322.966,75

